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Probability

Consider an experiment whose outcome isn't predictable with certainty beforehand. Although the outcome will not be known beforehand, let us suppose that the set of all possible outcomes is.
This set of all possible outcomes of an experiment is known as the sample space and is denoted by S.
Such an experiment is called a random experiment.

Any subset E of the sample space is known as an event.
An event is a set consisting of possible outcomes of an experiment.

If the outcome of the experiment is contained in E, then E has occurred.

E.g. There are 2 children in a house.
Random experiment: we check the gender of the 2 children.
Sample space: S = { FF, FM, MF, MM }
Event E: both are of different sexes: E = { FM, MF } (say order matters)

E.g. 2 die are rolled, we see the number on both.
Sample space: S = { (1 1), (1 2), .., (6 6) }
E1: Sum of both the numbers is prime:
E1 = {
        (1 1), (1 2), (1 4), (1 6),
        (2 1), (2 3), (2 5),
        (3 2), (3 4),
        (4 1), (4 3),
        (5 2), (5 6),
        (6 1), (6 5)
    }
|E1| = 15

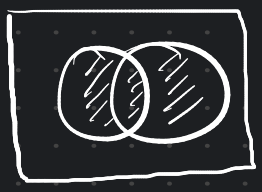
Probability: a measure of the likelihood of an event to occur.
Probability of an event E occurring:

P(E) = # of favorable outcomes in E / Total # of outcomes possible in S

Q. There are 6 pillows in a bed. 3 are red, 2 are yellow, and 1 is blue. You pick one. What is the prob that it is yellow?

2/(3+2+1) = 1/3

Union of events:
For any 2 events E and F of a sample space S, we define G = E U F, called the union of the events E and F, to consist of all outcomes that are either in E or in F or in both.



Mutually exhaustive events:
Events E1, E2, ..., are mutually exhaustive if their union is the sample space, i.e. at least one of them will always occur.
If E U F = S, then E and F are mutually exhaustive.

Intersection of events:
For any 2 events E and F, we define G = E ∩ F, called the intersection of the events E and F, to consist of all the outcomes that are in both E and F.
I.e. G would occur if both E and F occur.



$$|E \cup F| = |E| + |F| - |E \cap F|$$

Mutually exclusive events:
If $E \cap F = \phi$, implying that both of them cannot occur at the same time, they are called mutually exclusive.



Union and intersection are commutative.

$$E \cup F = F \cup E$$
$$E \cap F = F \cap E$$

Union and intersection are associative.

$$(E \cup F) \cup G = E \cup (F \cup G)$$
$$(E \cap F) \cap G = E \cap (F \cap G)$$

Union and intersection are distributive.

$$(E \cup F) \cap G = (E \cap G) \cup (F \cap G)$$

Complement of an event:
For any event E, we define the event E^c or $\bar{E}$, called the complement of E to consist of all outcomes in the sample space S that are not in E.
$\bar{E}$ will occur iff E doesn't.



$\bar{E} = S - E$

$|\bar{E}| = |S| - |E|$

divide by $|S|$

$P(\bar{E}) = 1 - P(E)$

De Morgan's Laws: $\overline{E \cup F} = \bar{E} \cap \bar{F}$

$\overline{E \cap F} = \bar{E} \cup \bar{F}$

Subsets:
For any 2 events E and F, if all outcomes in E are also in F, then we say that E is contained in F.

$E \subseteq F$ (or $F \supseteq E$)

Then the occurrence of E implies occurrence of F.

$E \subseteq F$ and $F \subseteq E \Rightarrow$ E & F are the same.

Axioms of Probability:

Axiom 1: $0 <= P(E) <= 1$
Axiom 2: $P(S) = 1$ (certain event)
Axiom 3: For any sequence of mutually exclusive events E1, E2, ...
(i.e. $Ei \cap Ej = \phi$ for all i, j)

$P(E1 \cup E2 \cup ... \cup En) = P(E1) + P(E2) + ... + P(En)$

$$P\left(\bigcup_{i=1}^{\infty} E_i\right) = \sum_{i=1}^{\infty} P(E_i)$$

Q. A total of 30% of American males smoke cigarettes, 10% smoke cigars, and 5% smoke both cigars and cigarettes. What % of American males smoke neither?

→ $n(\text{cigar or cigarettes}) = 30 + 10 - 5 = 35\%$

$\therefore n(\text{neither}) = 100 - 35 = 65\%$

Odds in favor:
Odds in favor of an event A $= \frac{P(A)}{P(\bar{A})} = \frac{P(A)}{1 - P(A)} = \frac{|A|}{|\bar{A}|}$

Odds against:
Odds against an event A $= \frac{P(\bar{A})}{P(A)} = \frac{1 - P(A)}{P(A)} = \frac{|\bar{A}|}{|A|}$

Q. Find the odds in favor and odds against throwing a die to get a 3.

→ $P(E) = 1/6$, $P(\bar{E}) = 5/6$

Odds in favor $(E) = 1/5$

Q. If odds in favor of Rahul solving a problem are 4:3 and the odds against Aman solving it are 2:6.
Find the probabilities of both of them solving the problem.

→ $\frac{P(R)}{1 - P(R)} = \frac{4}{3} \Rightarrow P(R) = \frac{4}{7}$

$\frac{P(\bar{A})}{1 - P(\bar{A})} = \frac{1}{3} \Rightarrow P(\bar{A}) = \frac{1}{4} \Rightarrow P(A) = 3/4$.

Conditional Probability

- suppose you roll a pair of dice.
The sample space has the following 36 outcomes:
{ (1, 1), (1, 2), ..., (6, 6) }

What is the prob. that the sum of #s on the 2 dice is 8?

→ $E = \{(2,6), (6,2), (3,5), (5,3), (4,4)\}$

$P(E) = 5/36$

Suppose we further observe that the first die lands a 3. Then given this information, what is the prob that the sum is 8?

→ $S = \{(3,1), (3,2), (3,3), \ldots, (3,6)\}$

$E = \{(3,5)\}$ $P(E) = 1/6$

or we could think of it as such

$|S| = 36$, A: first die lands a 3.

E: sum = 8

$$P(E|A) = \frac{|E \cap A|}{|A|} = \frac{|\{(3,5)\}|}{|A|}$$

Let E and F be 2 events. Then the conditional probability of event E, given that F has already happened is denoted by P(E|F).

$$P(E|F) = \frac{P(E \cap F)}{P(F)}$$

Q. On rolling a die, what's the probability of getting a prime # given that an odd # showed up?

→ If odd # showed up, $S = \{1, 3, 5\}$

$E = \{3, 5\}$ $P(E) = 2/3$.

Or A: prime, B: odd, $S = \{1, 2, 3, 4, 5, 6\}$

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{2/6}{3/6} = 2/3$$

Random expt: Rolling 2 dice together
E: both die show an odd #
F: sum of #s on both dice is 6

Q. P(E|F) = ?

$E \cap F = \{(1,5), (5,1), (3,3)\}$

$F = \{(1,5), (5,1), (2,4), (4,2), (3,3)\}$

$E = \{(1,1), (1,3), (1,5), (3,1) \ldots, (5,5)\}$

$n(E) = 3 * 3 = 9$

→ $P(E|F) = \frac{P(E \cap F)}{P(F)} = \frac{3/36}{5/36} = 3/5$

Q. P(F|E) = ?

→ $P(F|E) = \frac{P(F \cap E)}{P(E)} = 3/9$

Q. A bin contains 3 defective (that immediately fail when put in use),
5 partially defective (would fail in a couple of hours), and
10 acceptable transistors.
A transistor is chosen at random from the bin and put to use.
If it does not fail immediately, what is the prob that it is acceptable?

→ Given that it doesn't fail immediately, sample space is either partial/acceptable.

$$P(\text{acceptable}) = \frac{10}{10+5} = 2/3$$

A: acceptable B: immediately fails

we want $P(A|\bar{B}) = \frac{P(A \cap \bar{B})}{P(B)}$ $A \cap \bar{B} = A$

$= 10/15 = 2/3$

Q. Ms. Perez figures that there's a 30% chance that her company will set up a branch office in Phoenix. If it does, she's 60% certain that she'll be made the manager of this new operation.
What is the prob that she'll be the manager of the new operation?

→ $P(\text{set up}) = \frac{30}{100}$

$P(\text{manager} \mid \text{setup}) = \frac{60}{100}$

$\frac{P(\text{manager} \cap \text{setup})}{P(\text{setup})} = 60/100$

$P(\text{manager} \cap \text{setup}) = \frac{60}{100} \times \frac{30}{100} = \frac{18}{100}$

Q. A box contains 4 white and 3 red balls. 2 balls are randomly selected and removed from the box. Now given that the first ball removed is white, what is the probability that the second ball is red?

→ $P(2 \text{ red} \mid 1 \text{ white}) = \frac{3}{3+3} = 1/2$

$P(AB)$ means $P(A \cap B)$

We know that $P(A|B) = \frac{P(A \cap B)}{P(B)}$

$$\Rightarrow P(A \cap B) = P(A|B) \cdot P(B)$$

Also, $P(B|A) = \frac{P(A \cap B)}{P(A)}$

$$P(A \cap B) = P(A) \cdot P(B|A)$$

Only true if $P(A) \neq 0$

Multiplication theorem:

> $P(A \cap B) = P(A) \cdot P(B|A)$

> $P(A_1 \cap A_2 \cap \ldots \cap A_n) = P(A_1) \cdot P(A_2|A_1) \cdot P(A_3|A_1 A_2) \cdot \ldots \cdot P(A_n|A_1 A_2 \ldots A_n)$

Q. A bag contains 3 red and 2 blue balls. Pick balls without replacement. What is the prob of getting 1st red, then blue ball?

→ $P(R \cap B) = P(R) \cdot P(B|R)$

$= 3/5 \cdot 2/4 = 3/10$

E_1 :- First red ball — $P(E_1) = 3/5$

E_2 :- second blue ball — $P(E_2|E_1) = 2/4 = 1/2$

$P(E_1 \cap E_2) = P(E_1) \cdot P(E_2|E_1) = 3/5 \times 1/2 = 3/10$

What is the prob of getting 3 consecutive red balls?

→ $= P(R_1) \cdot P(R_2|R_1) \cdot P(R_3|R_1R_2)$

$= 3/5 \cdot 2/4 \cdot 1/3 = 1/10$

Q. There's a lock and a 10 keys. We try opening the lock using the keys one by one, until we are able to do so. We obv do not try the same keys again. What is the probability that the lock is opened in the 5th trial?

→ prob. that A_1 fails $= A_1 = 9/10$

$P(A_1) \cdot P(A_2|A_1) \cdot P(A_3|A_2A_1) \cdot P(A_4|A_3A_2A_1)$

$\cdot P(\overline{A_5}|A_4A_3A_2A_1)$

↳ prob. of 5th working

$= 9/10 \cdot 8/9 \cdot 7/8 \cdot 6/7 \cdot 1/6$

$= 1/10$

Q. A fair coin is tossed till a head appears for the first time. The prob that the # of required tosses is odd is __.

→ Head appears for 1st time on 1 toss $= 1/2$

2nd $= 1/2 \cdot 1/2$

3rd $= 1/2 \cdot 1/2 \cdot 1/2$

P(odd # of tosses for 1st head) = 1st try or 3rd try or 5th try ...

$= 1/2 + \underbrace{1/2 \cdot 1/2 \cdot 1/2}_{TTH} + 1/2^5 + \cdots$

$= \frac{1}{2} + \frac{1}{2^3} + \frac{1}{2^5} + \cdots \infty$

$= \frac{1/2}{1 - 1/2^2} = 1/2 \cdot 4/3 = 2/3$

Independent Events:
If P(E|F) = P(E), then E is said to be independent of F
(i.e. E is independent of F if knowledge that F occurs doesn't change the probability that E occurs).

$$P(E|F) = P(E)$$
$$\Rightarrow \frac{P(E \cap F)}{P(F)} = P(E)$$
$$\Rightarrow \boxed{P(E \cap F) = P(E) \cdot P(F)}$$

2 events E and F are said to be independent if prob of their intersection is equal to the product of their independent probabilities.
If E is independent of F, then F is independent of E as well.
Events that aren't independent are called dependent.

e.g. expt: tossing 2 coins
E1: getting a head on the 1st coin, E2: getting a head on the 2nd coin

S = { HH, HT, TH, TT }
E1 = { HH, HT }, E2 = { HH, TH }
E1 ∩ E2 = { HH }
P(E1 ∩ E2) = 1/4 P(E1)*P(E2) = 1/2 * 1/2 = 1/4

e.g. expt: rolling 2 dice
E1: getting a 2 on the first die,
E2: getting a sum of 5
E3: getting a sum of 7

$S = \{(1,1), (1,2) \ldots, (6,6)\}$

$E_1 = \{(2,1), (2,2), \ldots, (2,6)\}$ $P(E_1) = 6/36 = 1/6$

$E_2 = \{(1,4), (4,1), (2,3), (3,2)\}$ $P(E_2) = 4/36 = 1/9$

$E_3 = \{(1,6), (6,1), (2,5), (5,2), (3,4), (4,3)\}$ $P(E_3) = 6/36 = 1/6$

$E_1 \cap E_2 = \{(2,3)\}$, $E_2 \cap E_3 = \phi$, $E_1 \cap E_3 = \{(2,5)\}$

$P(E_1 \cap E_2) = 1/36$ $P(E_2 \cap E_3) = 0$ $P(E_1 \cap E_3) = 1/36$

only $P(E_1 \cap E_3) = P(E_1) \cdot P(E_3)$
$1/36 = 1/6 \cdot 1/6$

E1 and E3 are independent events.
E1 and E2 are dependent.
E2 and E3 are dependent.

An urn contains 3 green and 2 red balls. You take out 2 balls with replacement. What is the probability of
- getting green and then red ball?
- getting both green balls?

→ P(getting green then red) = $3/5 \cdot 2/5 = 6/25$

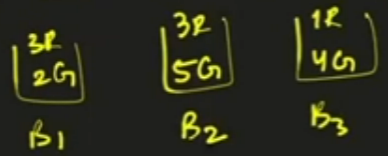
P(getting both green) = $3/5 \cdot 3/5 = 9/25$

↳ $P(G_1) \cdot P(R_2|G_1)$

$P(G_1) \cdot P(R_2)$

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eg There are three bags containing



If we draw 1 ball from each bag. what is the probability of:

(i) Getting a green, red and green balls from first, second & 3rd bag?

→ $2/5 \cdot 3/8 \cdot 4/5 = 3/25$

(ii) Getting 2 green and 1 red balls?

→ GGR + GRG + RGG

$= \frac{2}{5} \cdot \frac{5}{8} \cdot \frac{1}{5} + \frac{2}{5} \cdot \frac{3}{8} \cdot \frac{4}{5} + \frac{3}{5} \cdot \frac{5}{8} \cdot \frac{4}{5}$

$= \frac{10+24+60}{5 \cdot 8 \cdot 5} = \frac{5+12+30}{100} = \frac{47}{100}$

—x—

eg: A card is selected at random from an ordinary deck of 52 playing cards. If A is event that the selected card is an ace and H is the event that it is a heart. Are A and H independent?

→ $P(A) = 4/52$ $P(H) = 13/52$ ∴ yes.

$P(A \cap H) = 1/52$

$P(A) \cdot P(H) = \frac{4}{52} \cdot \frac{13}{52} = 1/52$

If A and B are independent, then
> A and B' are independent
> A' and B are independent
> A' and B' are independent

Proof:

i) $A \cap \bar{B} = A - (A \cap B)$

$\Rightarrow \frac{|A \cap \bar{B}|}{|S|} = \frac{|A|}{|S|} - \frac{|A \cap B|}{|S|}$

$P(A \cap \bar{B}) = P(A) - P(A) \cdot P(B)$

$P(A \cap \bar{B}) = P(A)[1 - P(\bar{B})]$

$P(A \cap \bar{B}) = P(A) \cdot P(\bar{B})$ $\therefore$ <i>

ii) using symmetry

iii) $P(\bar{A} \cap \bar{B}) = P(\overline{A \cup B})$

$= 1 - P(A \cup B)$

$= 1 - [P(A) + P(B) - P(A \cap B)]$

$= 1 - P(A) - P(B) + P(A) \cdot P(B)$

$= [1 - P(A)][1 - P(B)]$

$= P(\bar{A}) \cdot P(\bar{B})$

Q. If 2 people A and B are interviewing for a job and the prob of their selections are P(A) = 1/2 and P(B) = 1/3.
What is the probability that both of them are selected?

→ $P(A \cap B) = P(A) \cdot P(B) = \frac{1}{2} \cdot \frac{1}{3} = \frac{1}{6}$.

What is the probability that exactly one of them is selected?

→ $P(A \cap \bar{B}) + P(\bar{A} \cap B) = \frac{1}{2} \cdot \frac{2}{3} + \frac{1}{2} \cdot \frac{1}{3}$

$= \frac{1}{3} + \frac{1}{6} = \frac{1}{2}$

Prove that A is independent of all events if
i. P(A) = 1 (a certain event is independent of all events)
ii. P(A) = 0

→ $P(A) = 1 \quad \therefore A = S$

Any event B will be subset $B \subseteq A$.

$P(A \cap B) = P(B) = P(B) \cdot 1$

$= P(B) \cdot P(A)$

→ ii) $P(A) = 0$

$P(A) \cdot P(B) = 0$

$A \cap B = \phi$

$P(A \cap B) = 0$

Mutually exclusive events with non-zero probabilities are always dependent.

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GATE CS 2003

Let P(E) denote the probability of the event E. Given that P(A) = 1, P(B) = 1/2. the values of P(A|B) and P(B|A) respectively are :-

A) 1/4, 1/2

B) 1/2, 1/4

C) 1/2, 1

D) 1, 1/2 ✓

→ A is certain. ∴ It is independent of any event.

GATE CS 2000

E_1 and E_2 are two events in a probability space satisfying the following constraints:

$Pr(E_1) = Pr(E_2)$

$Pr(E_1 \cup E_2) = 1$

E_1 and E_2 are independent.

The value of $Pr(E_1)$, the probability of event E_1 is

A) 0

B) 1/4

C) 1/2

D) 1

→ $P(E_1 \cup E_2) = P(E_1) + P(E_2) - P(E_1 \cap E_2)$

$1 = x + x - x \cdot x$

$x^2 - 2x + 1 = 0$

$(x-1)^2 = 0 \Rightarrow x = 1$

Independence of 3 events:
3 events A, B, and C are independent if all the following hold:

$P(A \cap B) = P(A) \cdot P(B)$

$P(B \cap C) = P(B) \cdot P(C)$

$P(A \cap C) = P(A) \cdot P(C)$

$P(A \cap B \cap C) = P(A) \cdot P(B) \cdot P(C)$

we check 2 at a time, 3 at a time, ..., n at a time.

Q. A ball is drawn from a bag containing 4 balls #d 1, 2, 3, 4 with each outcome equally likely.
E: getting a 1 or a 2 { 1, 2 }
F: getting a 1 or a 3 { 1, 3 }
G: getting a 1 or a 4 { 1, 4 }

→ $P(E) = P(F) = P(G) = 2/4 = 1/2$

$P(E \cap F) = 1/4$ $P(E \cap G) = 1/4$ $P(F \cap G) = 1/4$

so they are pairwise independent.

E & F are independent, F & G are ind, E & G are.

But $P(E \cap F \cap G) = 1/4$

$P(E) \cdot P(F) \cdot P(G) = 1/2 \cdot 1/2 \cdot 1/2 = 1/8$.

∴ All 3 are NOT independent of each other.

If A, B, and C are independent, prove that these events are also independent:

i) $A \cup B$ and C

ii) $A' \cup B$ and C

iii) $A' \cup B'$ and C

iv) A', B', and C'

$\rightarrow P((A \cup B) \cap C) = P((A \cap C) \cup (B \cap C))$

$= P(A \cap C) + P(B \cap C) - P(A \cap C \cap B \cap C)$

$= P(A \cap C) + P(B \cap C) - P(A \cap B \cap C)$

$= P(A)P(C) + P(B)P(C) - P(A)P(B)P(C)$

$= P(C) \cdot [P(A) + P(B) - P(A)P(B)]$

$= P(C) \cdot P(A \cup B)$

$\therefore A \cup B$ and C are independent.

Why we can prove A and B∪C are independ
B and A∪C a

$\rightarrow P((A' \cup B) \cap C) = P((A' \cap C) \cup (B \cap C))$

$= P(A' \cap C) + P(B \cap C) - P(A' \cap C \cap B \cap C)$

$= P(A')P(C) + P(B)P(C) - P(A')P(B)P(C)$

$= P(C)[P(A') + P(B) - P(A')P(B)]$

$= P(C) \cdot P(A' \cup B)$

∴ A & B are independent, A' & B' are ind., A' & B are ind., A & B' are ind. ∴

$\rightarrow P((A' \cup B') \cap C) = P((A' \cap C) \cup (B' \cap C))$

$= P(A' \cap C) + P(B' \cap C) - P(A' \cap C \cap B' \cap C)$

$= P(A')P(C) + P(B')P(C) - P(A')P(B')P(C)$

$= P(C)[P(A') + P(B') - P(A')P(B')]$

$= P(C) \cdot P(A' \cup B')$

$\therefore$

→ A', B' and C'

$$P(A' \cap B') = 1 - P(A \cup B)$$
$$= 1 - P(A) - P(B) + P(A \cap B)$$
$$= P(A \cap B) - P(A) - P(B) + 1$$
$$= P(A) \cdot P(B) - P(A) - P(B) + 1$$
$$= P(A)[P(B) - 1] - 1[P(B) - 1]$$
$$= (1 - P(A))(1 - P(B))$$

∴ A' & B' are ind.,
by. B' & C' are ind,
A' & C' are ind. } pairwise

$$P(A' \cap B' \cap C') = 1 - P(A \cup B \cup C)$$
$$= 1 - [P(A \cup B) + P(C) - P((A \cup B) \cap C)]$$
$$= 1 - P(A \cup B) - P(C) + P(A \cup B) \cdot P(C)$$
$$= 1 - P(A) - P(B) + P(A) \cdot P(B)$$
$$- P(C)$$
$$+ P(A) \cdot P(C) + P(B) P(C) - P(A) P(B) P(C)$$
$$= [1 - P(A)][1 - P(B)][1 - P(C)]$$
$$= P(A') P(B') P(C').$$

Q. If P(A) = 1/2, P(B) = 1/3, P(C) = 1/4, and A, B, C are independent, find:

i) $P((A \cup B') \cap C)$

ii) $P(A' \cap B' \cap C')$

→ $P((A \cup B') \cap C) = P(A \cup B') \cdot P(C)$

$= [P(A) + P(B') - P(A)P(B')] \cdot P(C)$

$= \left[\frac{1}{2} + \frac{2}{3} - \frac{1}{2} \cdot \frac{2}{3}\right] \frac{1}{4}$

$= \left[\frac{1}{2} + \frac{1}{3}\right] \frac{1}{4} = \frac{5}{24}$

→ $P(A' \cap B' \cap C') = P(A') \cdot P(B') \cdot P(C')$

$= 1/2 \cdot 2/3 \cdot 3/4 = 1/4$

Random Variable:
In probability, a real-valued function defined over the sample space of a random experiment is called a random variable.

E.g. random experiment: throwing a die
S = { 1, 2, 3, 4, 5, 6 }
X: # that appears on the top of the die. X is a random variable.
P(X=1) = 1/6, P(X=2) = 1/6, ...

E.g. random experiment: throwing a pair of dice
random variable X: sum of the #s that appear on the top of the dice.

Probability of 1 random variable:
The probability of a specific event A for a random variable X is denoted as P(X=A) or simply P(A).
Probability is calculated as the # of desired outcomes divided by the total # of outcomes.

For any random variable, sum of the probabilities of all outcomes in the sample space = 1.

Probability of multiple random variables:
We are more likely to work with many random variables. We assume that the 2 variables are related or dependent in some way.

Joint Probability: Probability of A and B.e

Marginal Probability: Probability of event X=A. (other random variables don't matter)

Conditional Probability: Probability of event A given event B.

E.g. random experiment: throwing a die and tossing a coin

X: result on die, possible values = { 1, 2, 3, 4, 5, 6 }
Y: result of toss, possible values { H, T }

Joint prob: refers to the likelihood of two events simultaneously
$P(A \cap B)$ or P(X=A and Y=B)

P(X=3 and Y=T) = 1/12 = P(X=3)*P(Y=T|X=3)
= P(X=3)*P(Y=T) = 1/6 * 1/2

Marginal prob: is the prob of a single event without considering the occurence of other events. It is obtained by summing the joint probabilities over the other events.

marginal prob of getting a 3:
P(X=3) = P(X=3, Y=H) + P(X=3, Y=T) = 1/12 + 1/12 = 1/6

lly, P(Y=H) = P(Y=H, X=1) + P(Y=H, X=6) + ... + P(Y=H, X=6) = 1/12 * 6 = 1/2

Conditional prob is the prob of 1 event happening given that another event has already occured.

The conditional prob of getting a tails given that the die shows a 4 is

P(Y=T | X=4) = P(Y=T) ..as here X and Y are independent
= 1/2

= P(Y=T, X=4)/P(X=4) = joint/marginal
= 1/12 / 1/6

Q. Consider 2 r.v.s X and Y with joint prob distribution function given in the table:

X ↓ \ Y →	2	4	5
1	1/12	1/24	1/24
2	1/6	1/12	1/8
3	1/4	1/8	1/12

1) $P(X=2, Y=5)$

2) $P(X \leq 2, Y \leq 4)$

3) $P(X=2)$

4) $P(Y=2 \mid X=1)$

5) Are x and y independent?

1)

X ↓ \ Y →	2	4	5
1	1/12	1/24	1/24
2	1/6	1/12	**1/8**
3	1/4	1/8	1/12

$= 1/8$

2)

X ↓ \ Y →	2	4	5
1	**1/12**	**1/24**	1/24
2	**1/6**	**1/12**	1/8
3	1/4	1/8	1/12

$= 9/24$

3)

X ↓ \ Y →	2	4	5
1	1/12	1/24	1/24
2	**1/6**	**1/12**	**1/8**
3	1/4	1/8	1/12

$= 3/8$

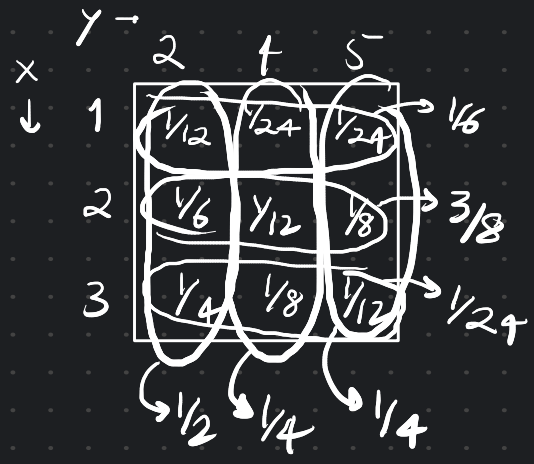
4)

X ↓ \ Y →	2	4	5
1	**1/12**	**1/24**	**1/24**
2	1/6	1/12	1/8
3	1/4	1/8	1/12

joint → $\frac{1/12}{1/12 + 1/24 + 1/24} = 1/2$

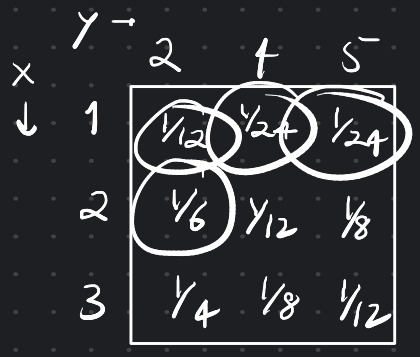
marginal →

⇒ we need to check for all possible values.



$P(X=1) = 1/6$
$X=2 \quad 3/8$
$X=3 \quad 11/24$
$Y=2 \quad 1/2$
$Y=4 \quad 1/4$
$Y=5 \quad 1/4$

X and Y will be independent
if $P(X=i, Y=j) = P(X=i) \cdot P(Y=j)$



$P(X=2, Y=2) = 1/6$

$P(X=2) \cdot P(Y=2) = 3/16$]

X and Y are not independent

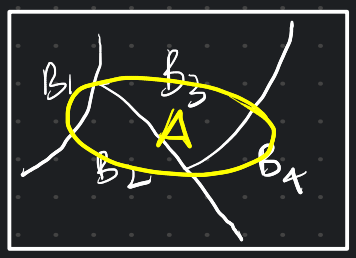
NO

Total Probability:
The of law of total probability is a fundamental rule relating marginal probabilities to conditional probabilities.
It expresses the total probability of an outcome which can be realized via several distinct events.

If { Bn, n = 1, 2, 3, ... } is the partition of the sample space (set of pairwise disjoint events whose union is the entire sample space),
Then for any event A of the same probability space:

$$P(A) = \sum_n P(A \cap B_n)$$

or $$P(A) = \sum_n P(A|B_n) \cdot P(B_n)$$



$$P(A) = P(A|B_1)\cdot P(B_1) + P(A|B_2)\cdot P(B_2) + P(A|B_3)\cdot P(B_3) + P(A|B_4)\cdot P(B_4)$$

as $A = (A \cap B_1) + (A \cap B_2) + (A \cap B_3) + (A \cap B_4)$

Say prob of it raining today is 60%. Prob of you going to college given it rains is 10%, and if it doesn't rain is 80%.
What is the prob of you going to college?

→ rain ∪ $\overline{\text{rain}}$ = total sample space (no other possibility)

$$P(\text{college}) = P(\text{college}|\text{rain})\cdot P(\text{rain}) + P(\text{college}|\text{not rain})\cdot P(\text{not rain})$$

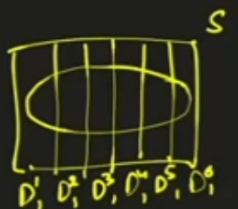
$$= \frac{10}{100}\cdot\frac{60}{100} + \frac{80}{100}\cdot\frac{40}{100} = 38\%$$

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eg: Experiment: Rolling 2 dice
What is the probability of getting 2 on second die?

Sol: $P(D_2^2) = P(D_1^1, D_2^2) + P(D_1^2, D_2^2) + P(D_1^3, D_2^2) + P(D_1^4, D_2^2) + P(D_1^5, D_2^2) + P(D_1^6, D_2^2)$

$= 1/36 \times 6 = 1/6$



Q. Suppose that two factories supply light bulbs. Factory X's bulbs work for over 5000 hours in 99% of all the cases, factory Y's in 95%.
Factory X supplies 60% of the total bulbs.
What is the chance that a random bulb will work for more than 5000 hrs?

$$\rightarrow P(\text{works}) = P(\text{works} \cap F_1) + P(\text{works} \cap F_2)$$
$$= P(\text{works} \mid F_1) \cdot P(F_1) + P(\text{works} \mid F_2) \cdot P(F_2)$$
$$= \frac{99}{100} \cdot \frac{60}{100} + \frac{95}{100} \cdot \frac{40}{100} = (59.4 + 38)\% = 97.4\%$$

Q. Bag A contains 2 green and 3 red balls. Bag B contains 3 red and 2 green balls.
You draw a ball from bag A and place it in bag B.
Now we draw a ball from bag B. What is the probability that it is red?

$$\rightarrow P(\text{ball from A is red}) = P(A_{\text{red}}) = 3/5$$

$$P(B_{\text{red}}) = \underbrace{\frac{4}{6} \cdot \frac{3}{5}}_{\text{Ball from A is red}} + \underbrace{\frac{3}{6} \cdot \frac{2}{5}}_{\text{Ball from A is not red.}} = \frac{2}{5} + \frac{1}{5} = \frac{3}{5}$$

Q. A bag contains 4 fair and 3 unfair coins. The prob of getting a heads on an unfair coin is 1/3. A coin is drawn randomly and tossed. Prob of getting a heads?

$$\rightarrow P(\text{heads}) = P(\text{heads} \mid \text{unfair}) \cdot P(\text{unfair}) + P(\text{heads} \mid \text{fair}) \cdot P(\text{fair})$$
$$= 1/3 \cdot 3/7 + 1/2 \cdot 4/7 = 1/7 + 2/7 = 3/7$$

Q. In a bag, there are 3 balls out of which one is red. The prob of any of the other balls being red is 0.50. What is the prob that a randomly chosen ball is red?



$$\rightarrow P(\text{red}) =$$
$$P(\text{red} \mid \text{ball 1}) \cdot P(\text{ball 1}) + P(\text{red} \mid b_2 \text{ or } b_3) \cdot P(b_2 \text{ or } b_3)$$
$$= 1 \cdot 1/3 + 1/2 \cdot 2/3 = 2/3$$

Q. There are 2 coins. The first gives a heads with prob 5/8, the second with 1/4. One of them is tossed. Prob of heads?

$$\rightarrow P(\text{heads}) = P(\text{heads} \mid \text{coin1})\,P(\text{coin1}) + P(\text{heads} \mid \text{coin2}) \cdot P(\text{coin2})$$
$$= \frac{5}{8} \cdot \frac{1}{2} + \frac{1}{4} \cdot \frac{1}{2} = \frac{7}{8} \cdot \frac{1}{2} = \frac{7}{16}$$

Bayes' Theorem:

Motivation: Say there are 3 bags. B1: { 2 red, 3 green balls },
B2: { 1 red, 4 green balls },
B3: { 3 red, 2 green balls }.
first we choose a bag, then pick a ball.

prob getting a red ball given bag B1 is chosen:
P(R|B1) = 2/5.

prob of getting a red ball and choosing bag B1?
P(R,B1) =

$$P(R \cap B_1) = P(B_1) \cdot P(R \mid B_1)$$
$$= \frac{1}{3} \cdot \frac{2}{5} = \frac{2}{15}$$

prob of getting a red ball?
P(R) =

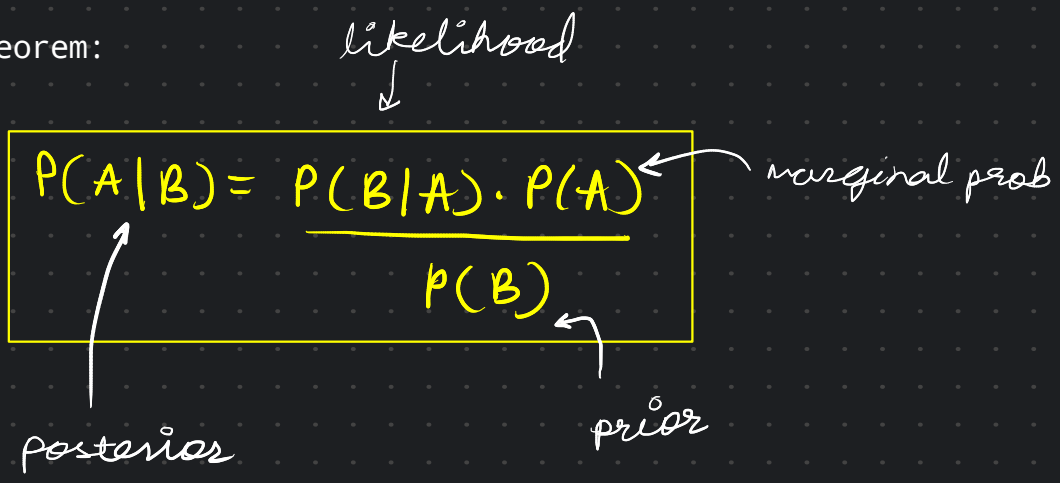
$$P(R) = P(R \mid B_1)\,P(B_1) + P(R \mid B_2) \cdot P(B_2) + P(R \mid B_3) \cdot P(B_3)$$
$$= \frac{2}{5} \cdot \frac{1}{3} + \frac{1}{5} \cdot \frac{1}{3} + \frac{3}{5} \cdot \frac{1}{3} = \frac{1}{3} \cdot \frac{6}{5} = \frac{2}{5}.$$

prob that B1 is chosen given that the ball drawn is red?
P(B1|R) = ??

$$P(R) = \underbrace{P(R \mid B_1)\,P(B_1)}_{\text{if a red ball is chosen, this is the contribution of bag } B_1} + P(R \mid B_2) \cdot P(B_2) + P(R \mid B_3) \cdot P(B_3)$$

$$P(B_1 \mid R) = \frac{\text{contribution of } B_1}{P(R)} = \frac{P(R \mid B_1)\,P(B_1)}{P(R)} = \frac{\frac{2}{5} \cdot \frac{1}{3}}{\frac{2}{5}} = \frac{1}{3}$$

Bayes' Theorem:



Q. Suppose that two factories supply light bulbs. Factory X's bulbs work for over 5000 hours in 99% of all the cases, factory Y's in 95%.
Factory X supplies 60% of the total bulbs.
What is the chance that a random bulb will work for more than 5000 hrs?

→ $P(\text{works}) = P(\text{works} \cap F_1) + P(\text{works} \cap F_2)$

$= P(\text{works}|F_1) \cdot P(F_1) + P(\text{works}|F_2) \cdot P(F_2)$

$= \frac{99}{100} \cdot \frac{60}{100} + \frac{95}{100} \cdot \frac{40}{100} = (59.4 + 38)\% = 97.4\%$

Given that the purchased bulb works for more than 5000 hrs, what is the prob that it was produced by factory X?

→ $P(F_1|\text{works}) = \frac{P(\text{works}|F_1)P(F_1)}{P(\text{works})} = \frac{\frac{99}{100} \cdot \frac{60}{100}}{\frac{97.4}{100}}$

$= 60 \cdot 9856\ \%$

Q. A person has undertaken a mining job. The probabilities of completion on time with and without rain are 0.42 and 0.90 respectively. If the prob that it'll rain is 0.45, what is the prob that it didn't rain, given that the mining job is done?

→ $P(M|R) = 0.42 \quad P(R) = 0.45$

$P(M|\bar{R}) = 0.9 \quad P(\bar{R}) = 0.55$

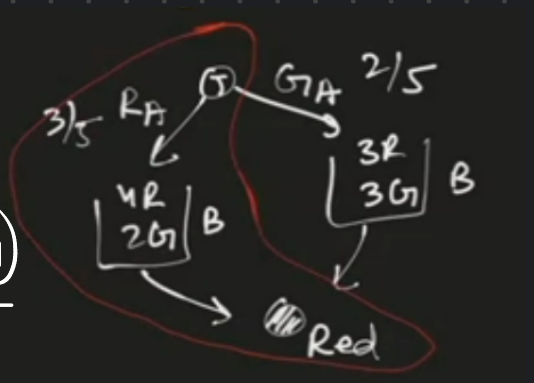
$P(\bar{R}|M) = \frac{P(M|\bar{R}) \cdot P(\bar{R})}{P(M)} = \frac{0.9 \times 0.55}{0.42 \times 0.45 + 0.9 \times 0.55} = 0.724$

Q. Bag A contains 2 green and 3 red balls. Bag B contains 3 red and 2 green balls. You draw a ball from bag A and place it in bag B.
Now we draw a ball from bag B. What is the probability that it is red?

→ $P(\text{ball from A is red}) = P(A_{red}) = 3/5$

$$P(B_{red}) = \frac{4}{6}\cdot\frac{3}{5} + \frac{3}{6}\cdot\frac{2}{5} = \frac{2}{5} + \frac{1}{5} = \frac{3}{5}$$

What is the prob of getting a red ball in the first draw (bag A), given that we got a red ball from the second draw (bag B)?



→ $P(B_{red}) = 3/5$

$$P(A_{red} \mid B_{red}) = \frac{P(B_{red} \mid A_{red}) \cdot P(A_{red})}{P(B_{red})}$$

$$= \frac{\frac{3}{5}\cdot\frac{4}{6}}{\frac{3}{5}\cdot\frac{4}{6} + \frac{2}{5}\cdot\frac{3}{6}}$$

$$= \frac{\frac{4}{6}}{\frac{4}{6} + \frac{2}{6}} = \frac{2}{3}$$

Q. In a bag, there are 3 balls out of which one is red. The prob of any of the other balls being red is 0.50. What is the prob that all the balls in the bag were red given that the randomly chosen ball is red?



N → not red

1 0.5 0.5

→

prob of each = 1/4

RRR RRN RNR RNN

$P(\text{Red}) = 1$ $P(\text{Red}) = 2/3$ $P(\text{red}) = 2/3$ $P(\text{Red}) = 1/3$

$$P(RRR \mid \text{Red}) = \frac{1\cdot\frac{1}{4}}{1\cdot\frac{1}{4} + \frac{2}{3}\cdot\frac{1}{4} + \frac{2}{3}\cdot\frac{1}{4} + \frac{1}{3}\cdot\frac{1}{4}} = \frac{1}{8/3} = 3/8$$

$$\frac{P(\text{Red} \mid RRR)\cdot P(RRR)}{P(\text{Red} \mid RRR)\cdot P(RRR) + P(\text{Red} \mid RNR)\cdot P(RNR) + P(\text{Red} \mid RRN)\cdot P(RRN) + P(\text{Red} \mid RNN)\cdot P(RNN)}$$

Q. Box P has 2 red and 3 blue balls. Box Q has 3 red and 1 blue ball.
A ball is selected as follows:
i. select a box
ii. choose a ball from the selected box such that each ball is equally likely.
The probs of selecting boxes P and Q are 1/3 and 2/3 resp.
Given that a ball selected from the above process is a red ball, the prob that it came from box P is ____.



→ $P(P|\text{red}) = \dfrac{P(\text{Red}|P)\cdot P(P)}{P(\text{Red}|P)\cdot P(P) + P(\text{Red}|Q)\cdot P(Q)}$

$= \dfrac{\frac{2}{5}\cdot\frac{1}{3}}{\frac{2}{5}\cdot\frac{1}{3} + \frac{3}{4}\cdot\frac{2}{3}} = 4/19$

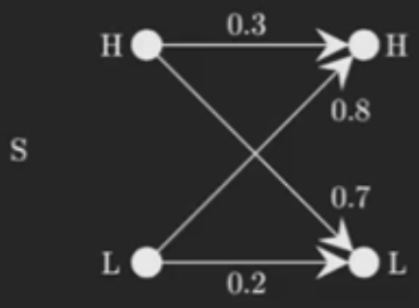
A bag contains 10 red balls and 15 blue balls. Two balls are drawn randomly without replacement. Given that the first ball drawn is red, the probability (rounded off to 3 decimal places) that both balls drawn are red is _____

→ 1st is red. Left: 9 red, 15 blue.

$P(\text{red}) = \dfrac{9}{24} = \dfrac{3}{8}$

A sender (S) transmits a signal, which can be one of the two kinds: H and L with probabilities 0.1 and 0.9 respectively, to a receiver (R).

In the graph below, the weight of edge (u, v) is the probability of receiving v when u is transmitted, where $u, v \in \{H, L\}$. For example, the probability that the received signal is L given the transmitted signal was H, is 0.7.



If the received signal is H, the probability that the transmitted signal was H (rounded to 2 decimal places) is _____.

→ $P(H_2|H_1) = 0.3$ $P(H_2|L) = 0.8$ $P(H_1) = 0.1$

$P(L_2|H_1) = 0.7$ $P(L_2|L) = 0.2$ $P(L_1) = 0.9$

$$P(H_1|H_2) = \frac{P(H_2|H_1)\,P(H_1)}{P(H_2)}$$

$$= \frac{P(H_2|H_1)\,P(H_1)}{P(H_2|H_1)\,P(H_1) + P(H_2|L_1)\cdot P(L_1)}$$

$$= \frac{(0.3)(0.1)}{(0.3)(0.1) + (0.8)(0.9)}$$

$$= \frac{1}{25}$$

A bag contains 4 fair and 3 unfair coins, P(Heads on any unfair coin) = ⅓.
A coin is drawn from the bag and tossed, what is the probability that it will land on heads?
Let's say you tossed a random coin from the bag and it landed on heads, what is the probability of you tossing a fair coin?

→ $P(f) = 4/7 \quad P(u) = 3/7$

$P(H|u) = 1/3$

$$P(f|H) = \frac{P(H|f) \cdot P(f)}{P(H)}$$

$$P(f|H) = \frac{1/2 \cdot 4/7}{1/3 \cdot 3/7 + 1/2 \cdot 4/7} = \frac{2/7}{\frac{1}{7}+\frac{2}{7}} = \frac{2}{3}$$

A fair coin is flipped twice, and it is given that at least one tail has been observed. The probability of getting two tails is

→ $S = \{HT, TH, TT\} \therefore P(TT) = 1/3$

A girl chooses a number uniformly at random from {1, 2, 3, 4, 5, 6}. If she chooses n, then she chooses another number uniformly at random from {1, . . . , n}. What is the probability that the second number is 5?

→ to get 2nd # 5, we need n at least 5.
choosing 1st # {5, 6}

$\{1,2,3,4,5\}$ $\{1,2,3,4,5,6\}$

$$P(5) = \frac{1}{6} \times \frac{1}{5} + \frac{1}{6} \times \frac{1}{6} = \frac{11}{180}$$